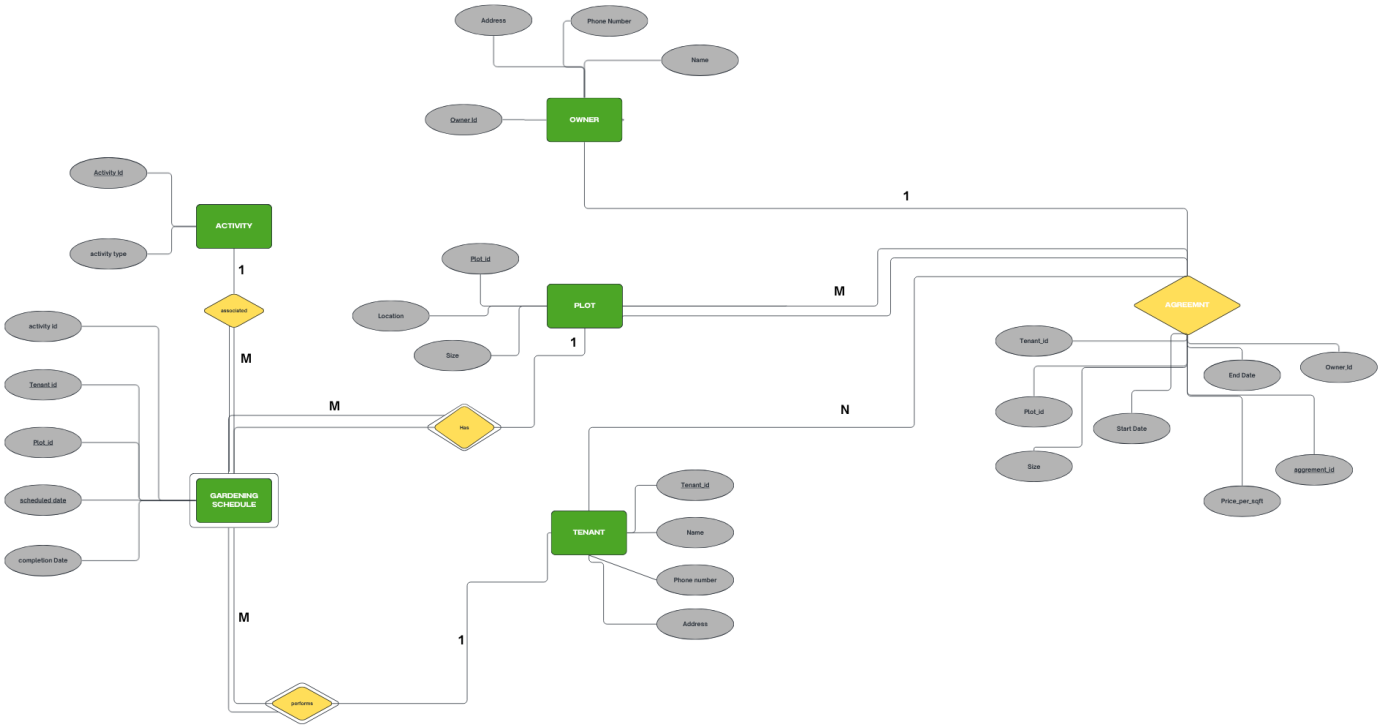
Assignment 2



Schema of GreenPatch Database

owners

name	aadhaar_number	street_address	city	state	pincode	phone

tenants

name	aadhaar_number	street_address	city	state	pincode	phone

plots

plot_id	location	size_sqft	owner_aadhaar

agreements

agreement_id	owner_aadhaar	tenant_aadhaar	plot_id	start_date	end_date	size_leased_sqft	price_per_sqft

gardening_schedules

schedule_id	activity_id	plot_id	tenant_aadhaar	schedule_date	completion_date	notes

activities

activity_id	activity_name

Relational Schema

owners
aadhaar_number CHAR(12) PRIMARY KEY,
name VARCHAR(100) NOT NULL,
street_address VARCHAR(255),
city VARCHAR(100),
state VARCHAR(100),
pincode CHAR(6),
phone CHAR(10)

tenants
aadhaar_number CHAR(12) PRIMARY KEY,
name VARCHAR(100) NOT NULL,
street_address VARCHAR(255),
city VARCHAR(100),
state VARCHAR(100),
pincode CHAR(6),
phone CHAR(10)

plots
plot_id INT AUTO_INCREMENT PRIMARY KEY,
location VARCHAR(150),
size_sqft DECIMAL(10,2),
owner_aadhaar CHAR(12),
FOREIGN KEY (owner_aadhaar) REFERENCES OWNERS (aadhaar_number)
ON UPDATE CASCADE
ON DELETE RESTRICT

agreements
agreement_id INT AUTO_INCREMENT PRIMARY KEY,
owner_aadhaar CHAR(12) NOT NULL,
tenant_aadhaar CHAR(12) NOT NULL,
plot_id INT NOT NULL,
start_date DATE NOT NULL,
end_date DATE,
size_leased_sqft DECIMAL(10,2),
price_per_sqft DECIMAL(8,2),
FOREIGN KEY (owner_aadhaar) REFERENCES OWNERS(aadhaar_number)
ON UPDATE CASCADE
ON DELETE RESTRICT,
FOREIGN KEY (tenant_aadhaar) REFERENCES TENANTS(aadhaar_number)
ON UPDATE CASCADE
ON DELETE RESTRICT
FOREIGN KEY (plot_id) REFERENCES PLOTS(plot_id)

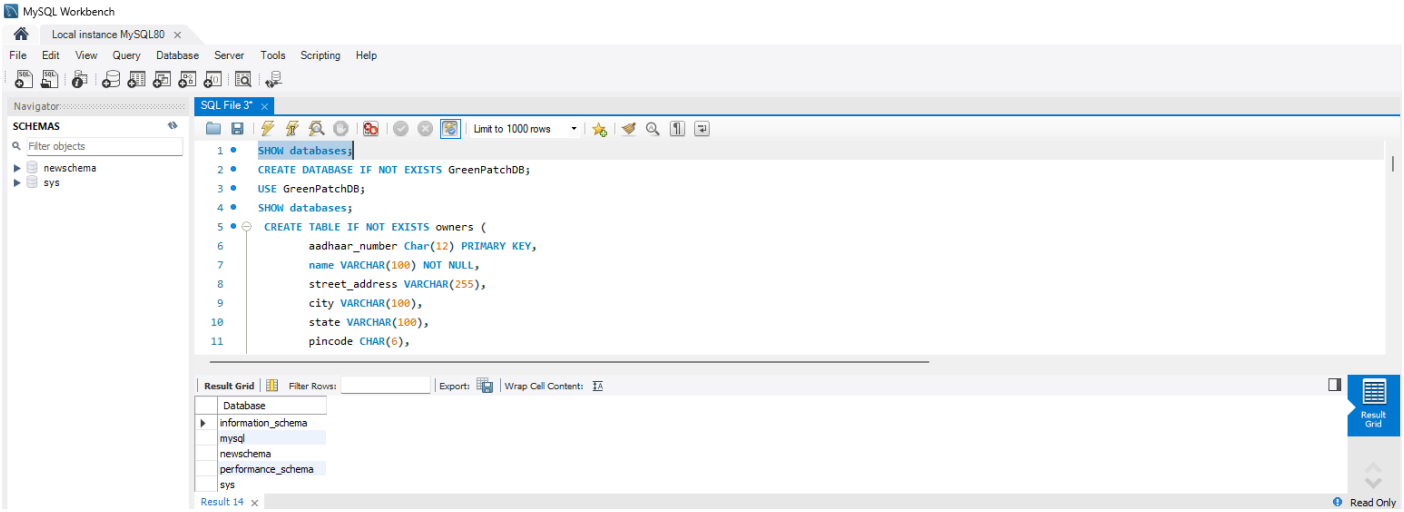
ON UPDATE CASCADE
ON DELETE RESTRICT

Gardening_schedules

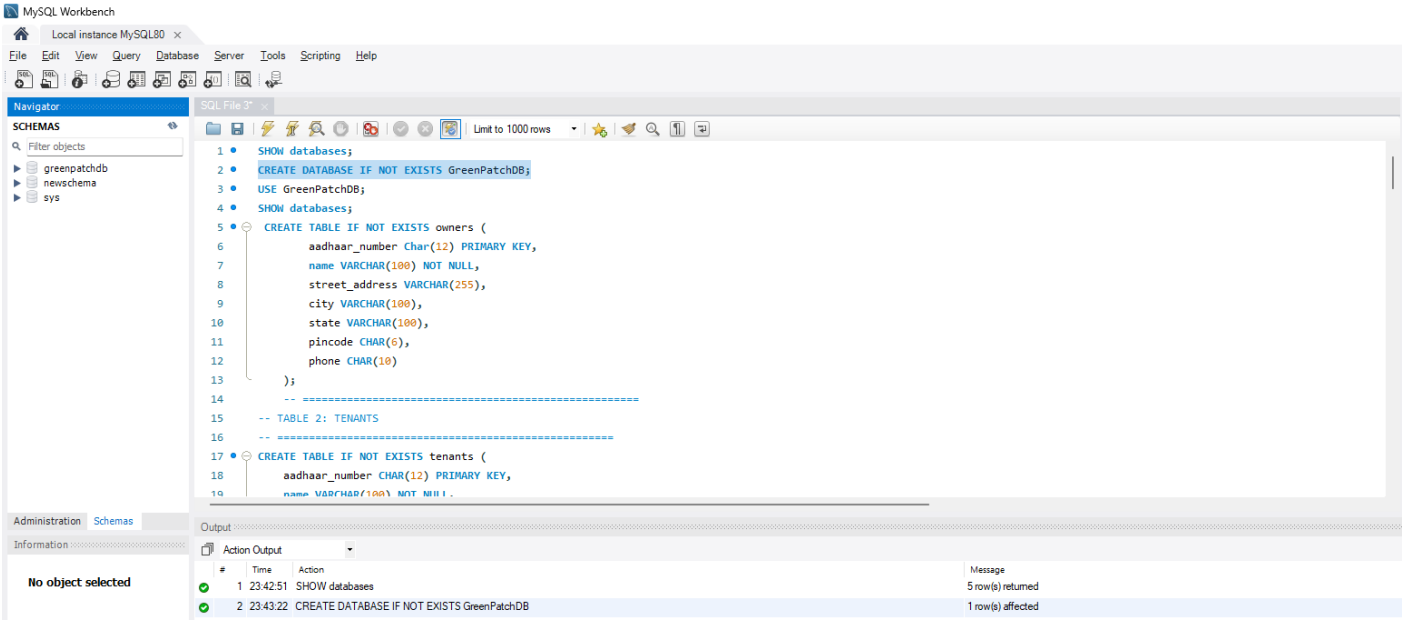
schedule_id INT AUTO_INCREMENT PRIMARY KEY,
activity_id INT NOT NULL,
plot_id INT NOT NULL,
tenant_aadhaar CHAR(12) NOT NULL,
scheduled_date DATE NOT NULL,
completion_date DATE,
notes VARCHAR(255),
FOREIGN KEY (activity_id) REFERENCES ACTIVITIES(activity_id)
ON UPDATE CASCADE
ON DELETE RESTRICT
FOREIGN KEY (plot_id) REFERENCES PLOTS(plot_id)
ON UPDATE CASCADE
ON DELETE RESTRICT
FOREIGN KEY (tenant_aadhaar) REFERENCES TENANTS(aadhaar_number)
ON UPDATE CASCADE
ON DELETE RESTRICT

activities
activity_id INT AUTO_INCREMENT PRIMARY KEY,
activity_name VARCHAR(100) NOT NULL

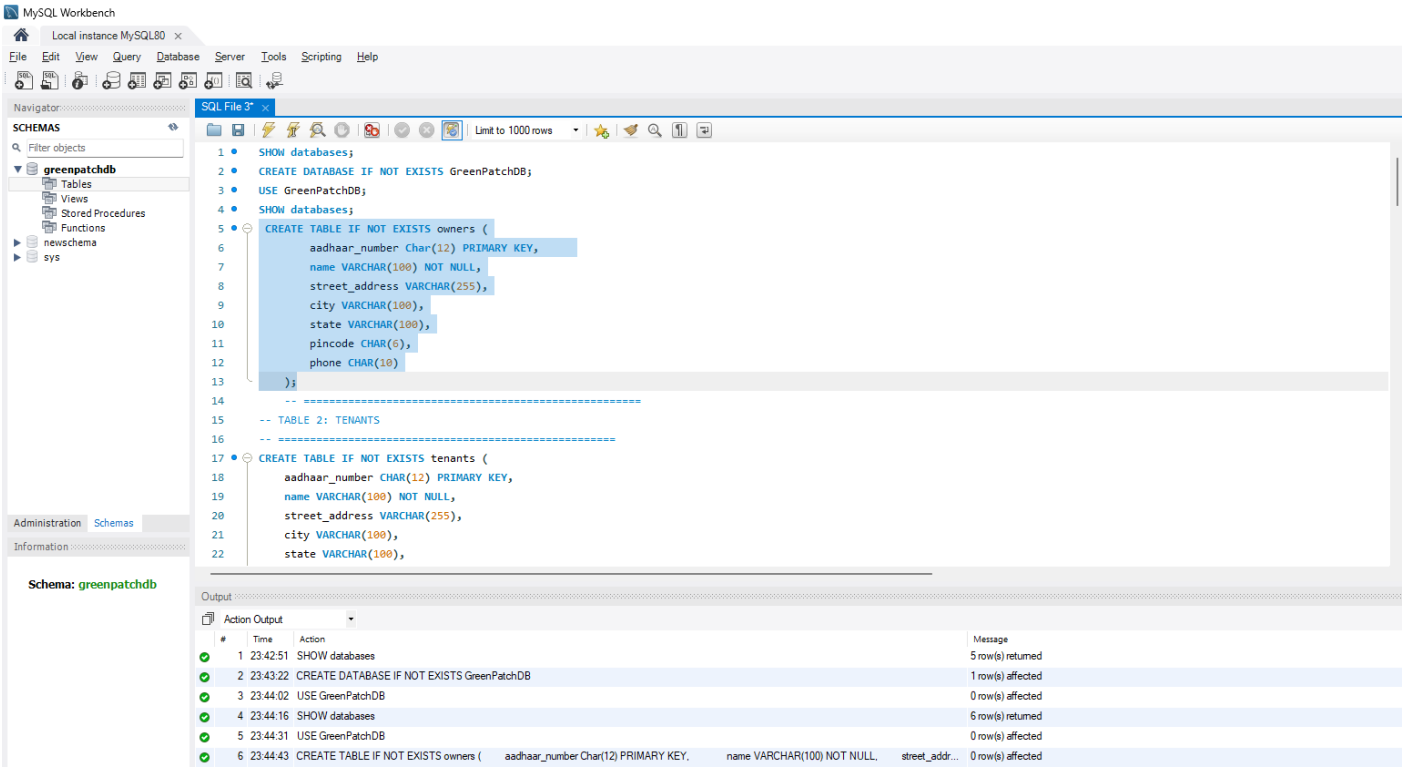
Show Databases



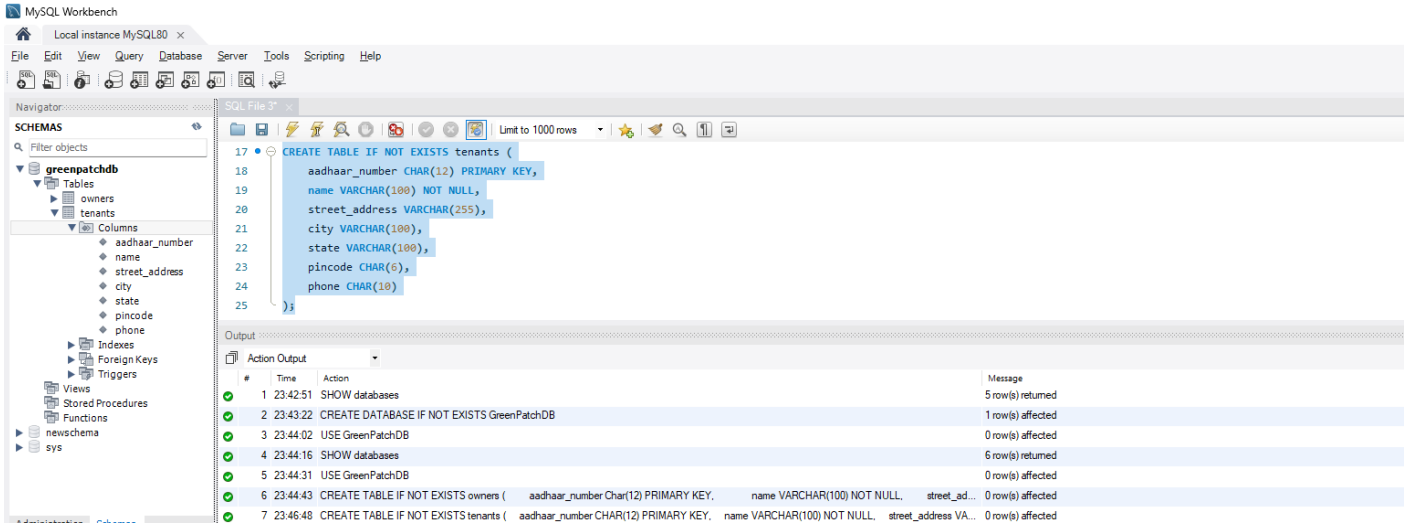
Creating GreenPatchDB



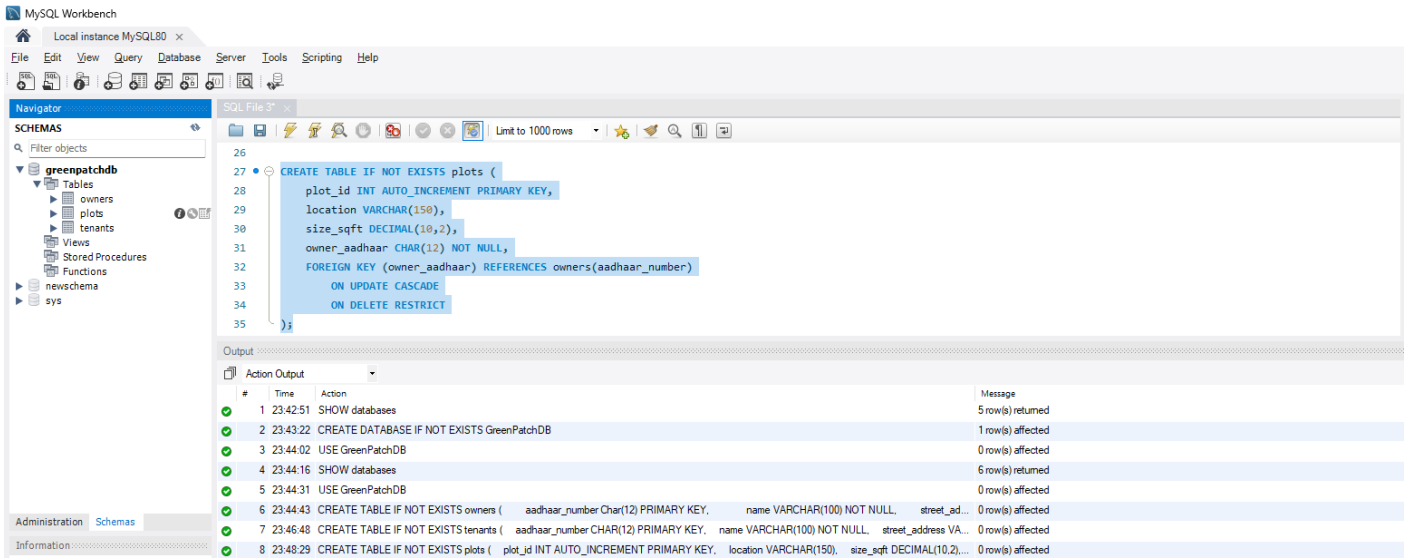
Creating owners table



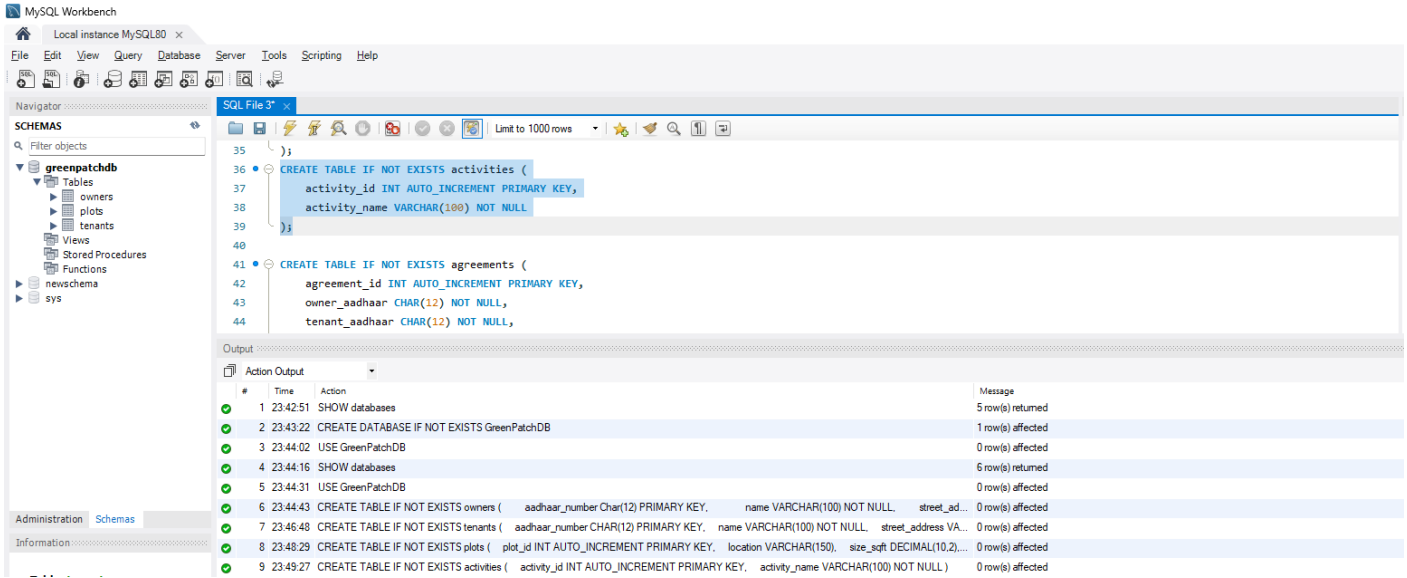
Creating tenant table



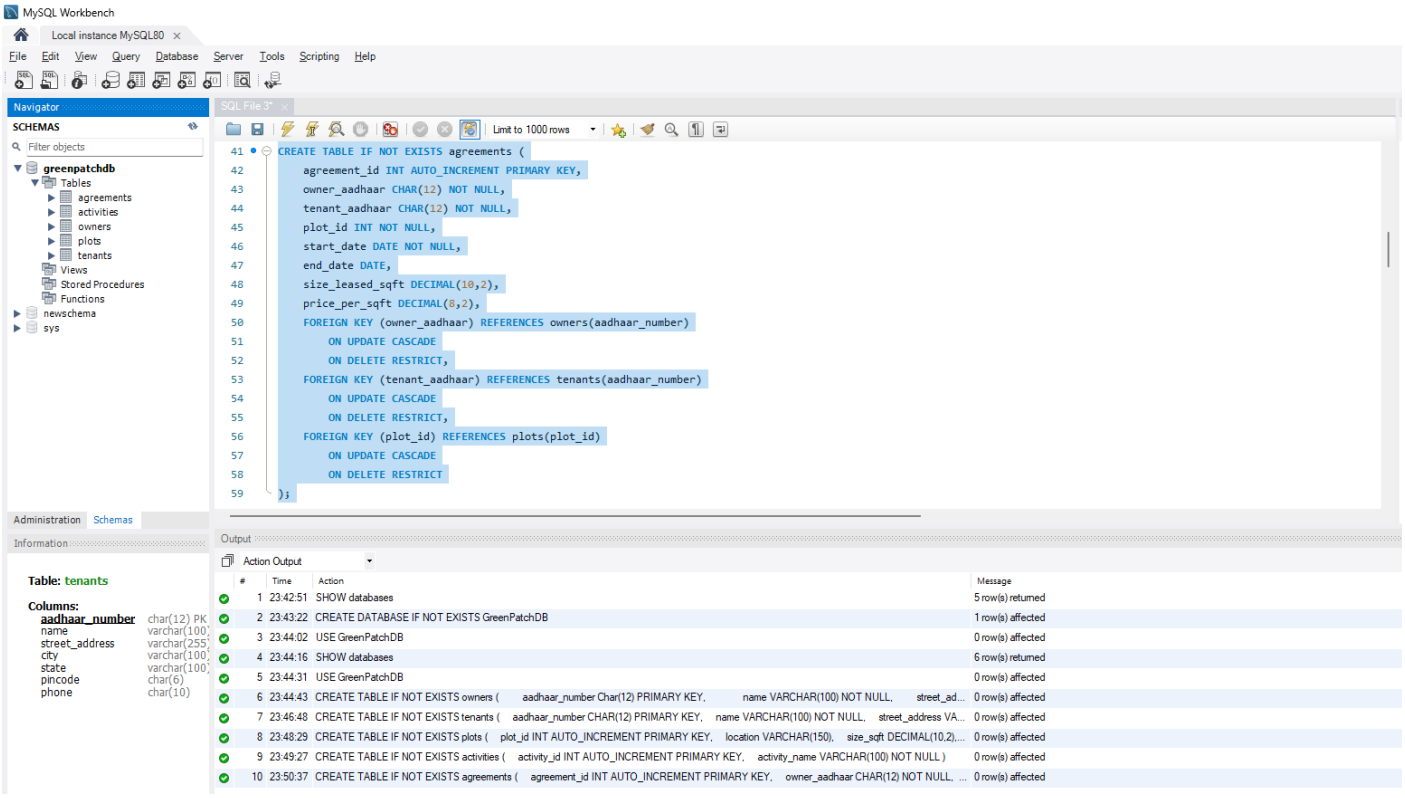
Creating plots table



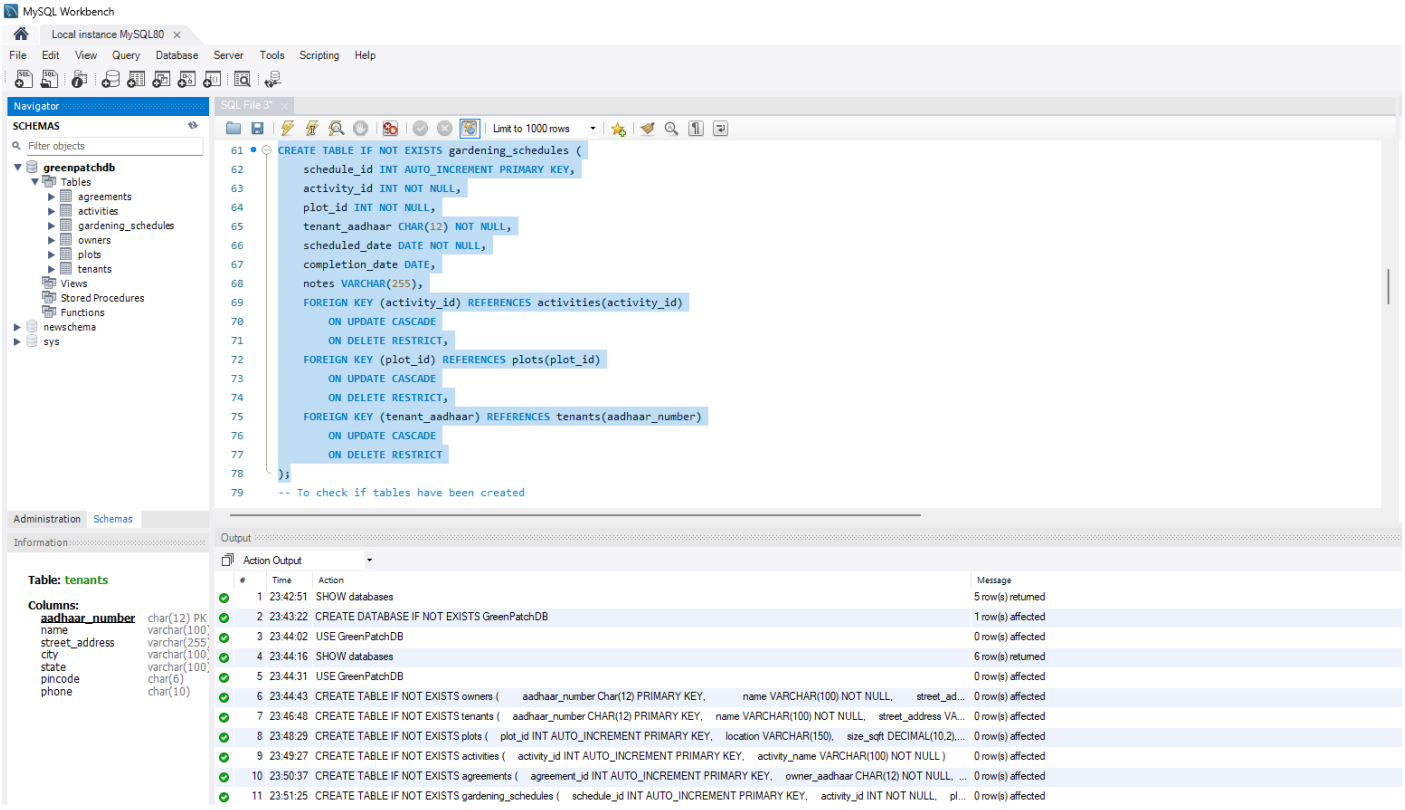
Creating activity table



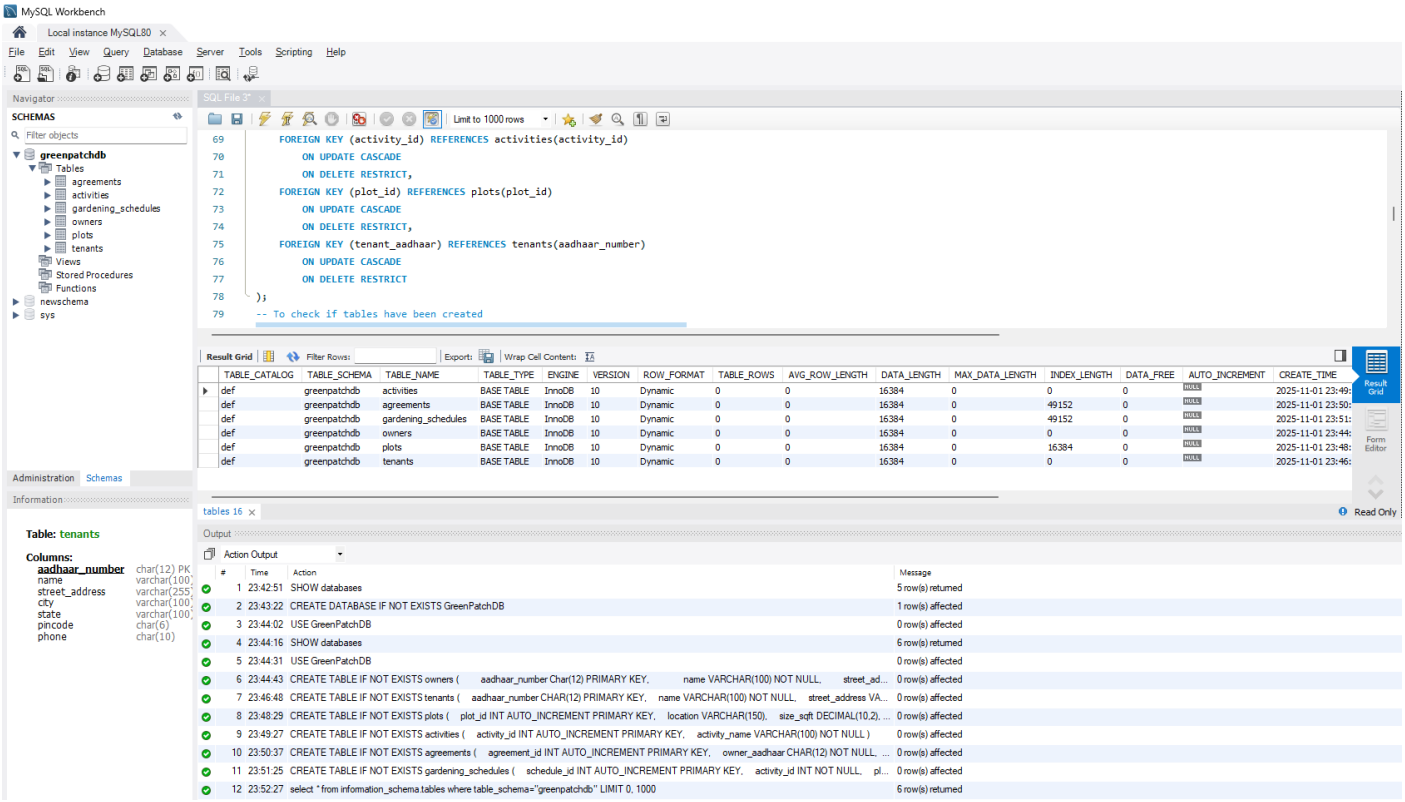
Creating agreements table



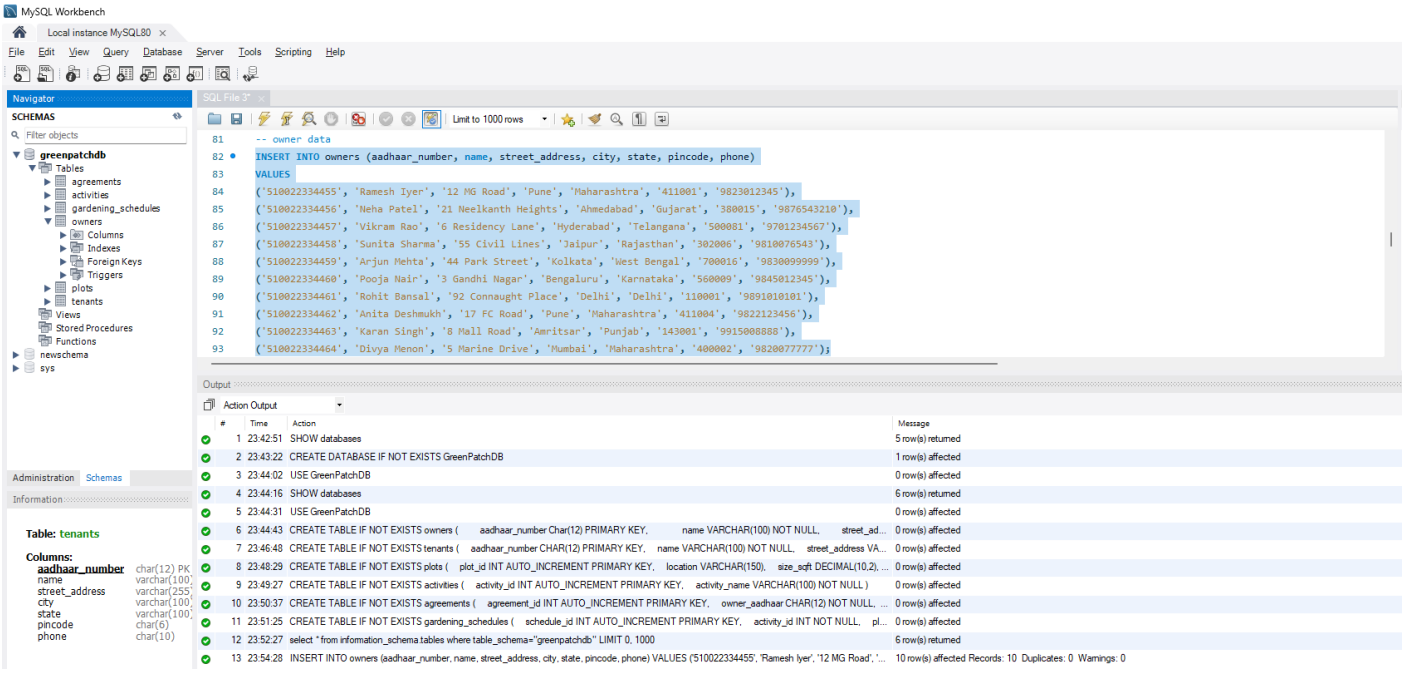
Creating gardening_schedules table



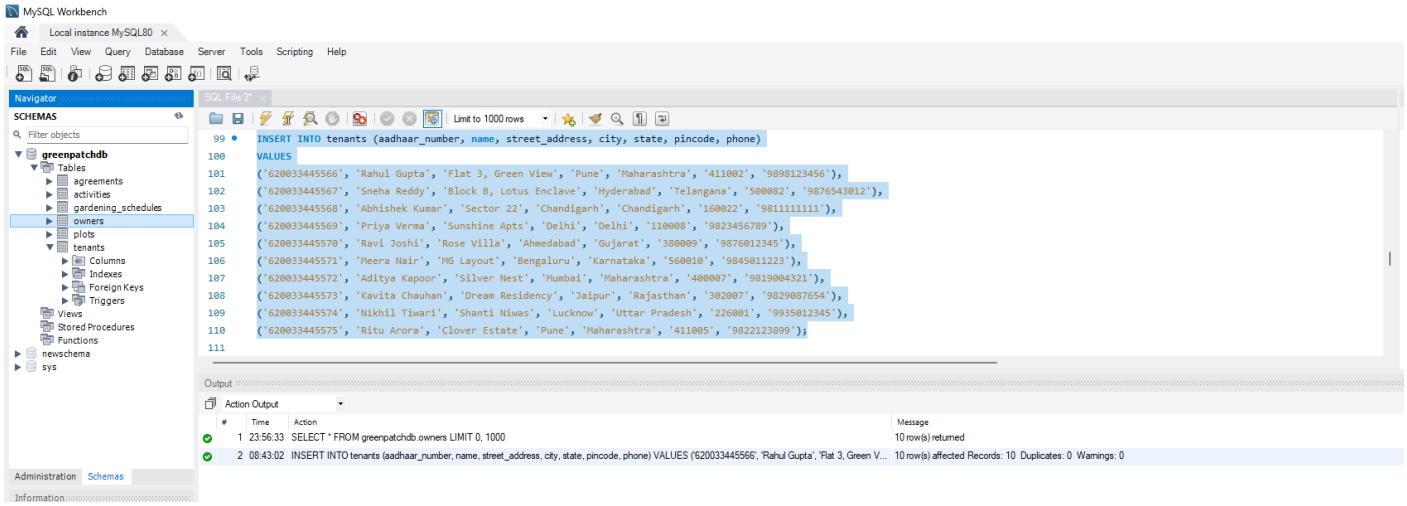
Checking tables



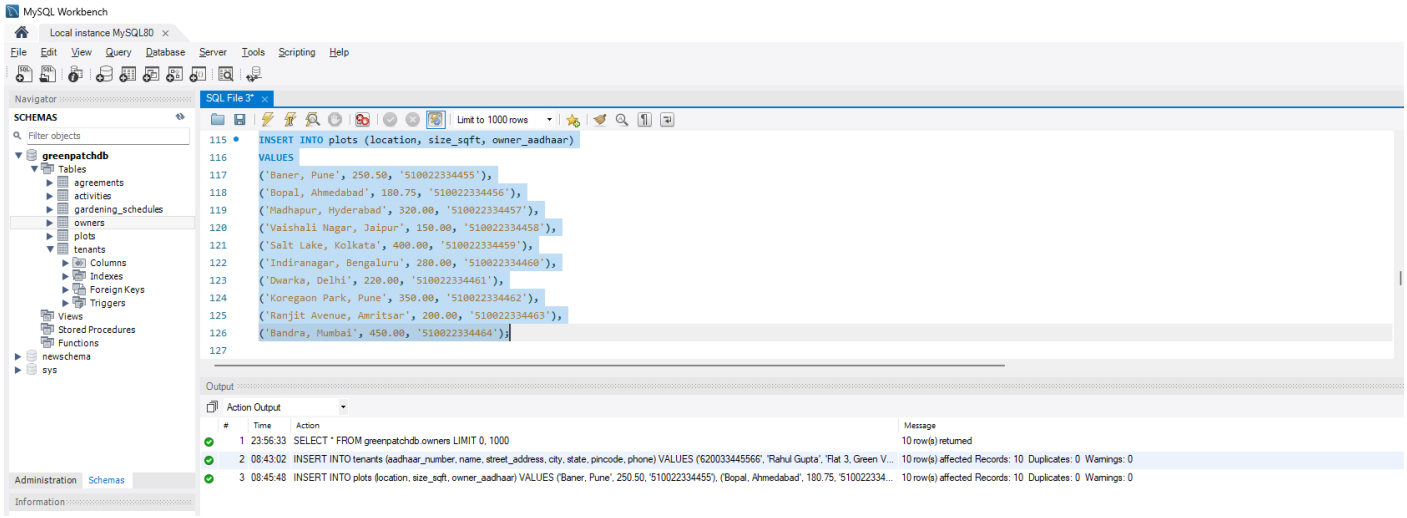
Inserting Data to owner table



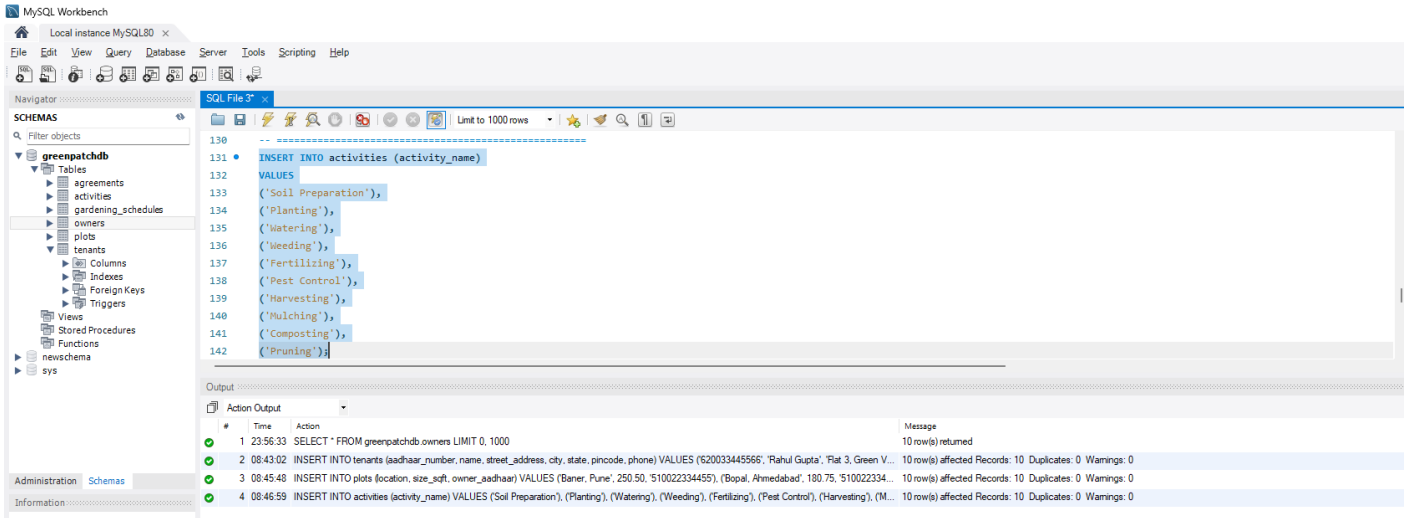
Inserting Data to tenants table



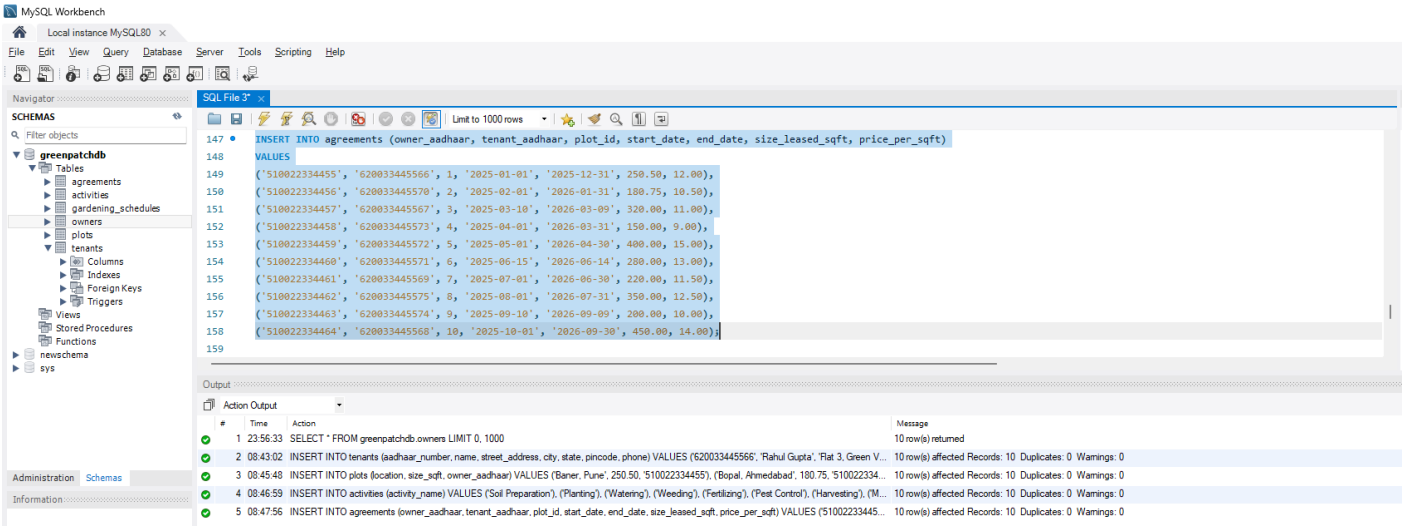
Inserting Data to plots table



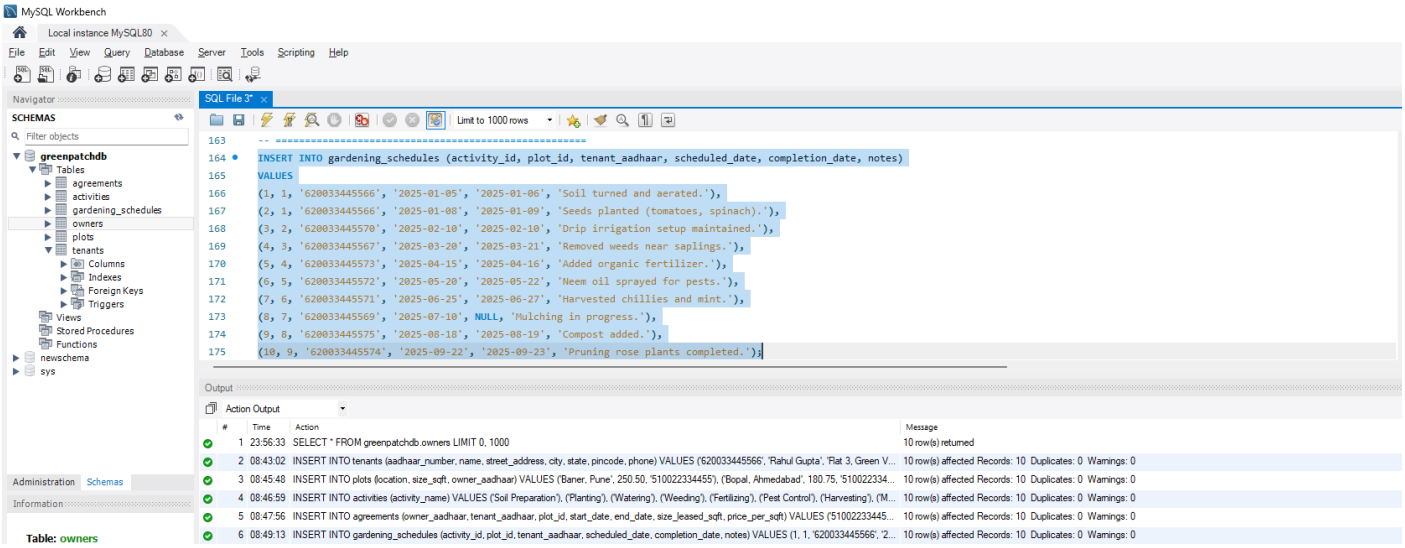
Insert Data to activities



Insert Data to agrrements

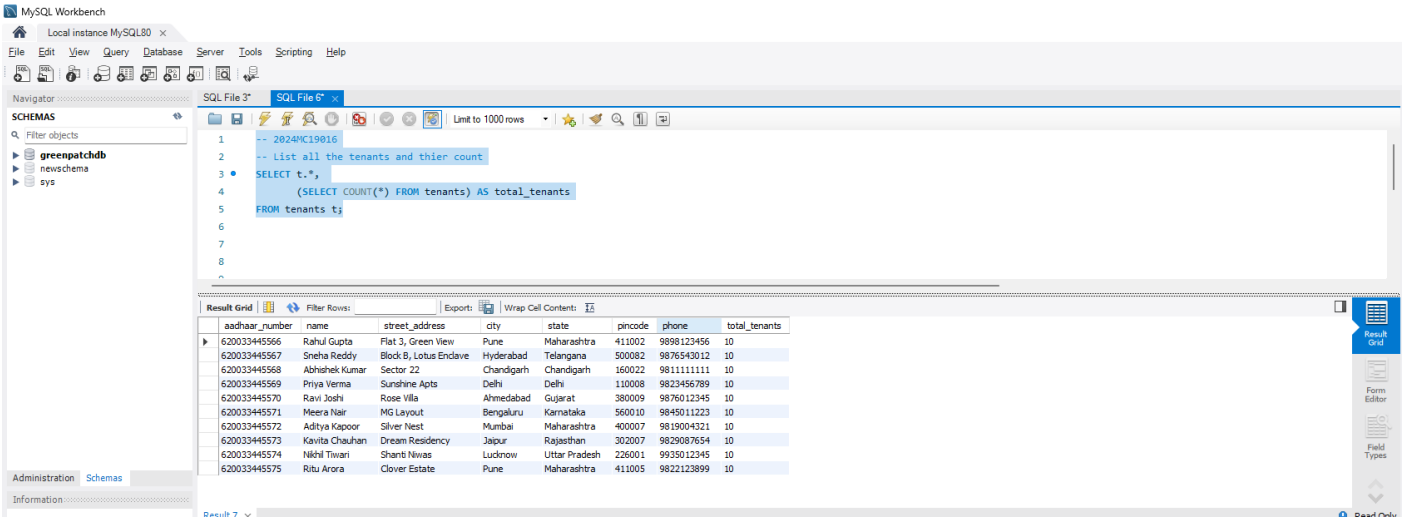


Insert Data to garndening_schedules

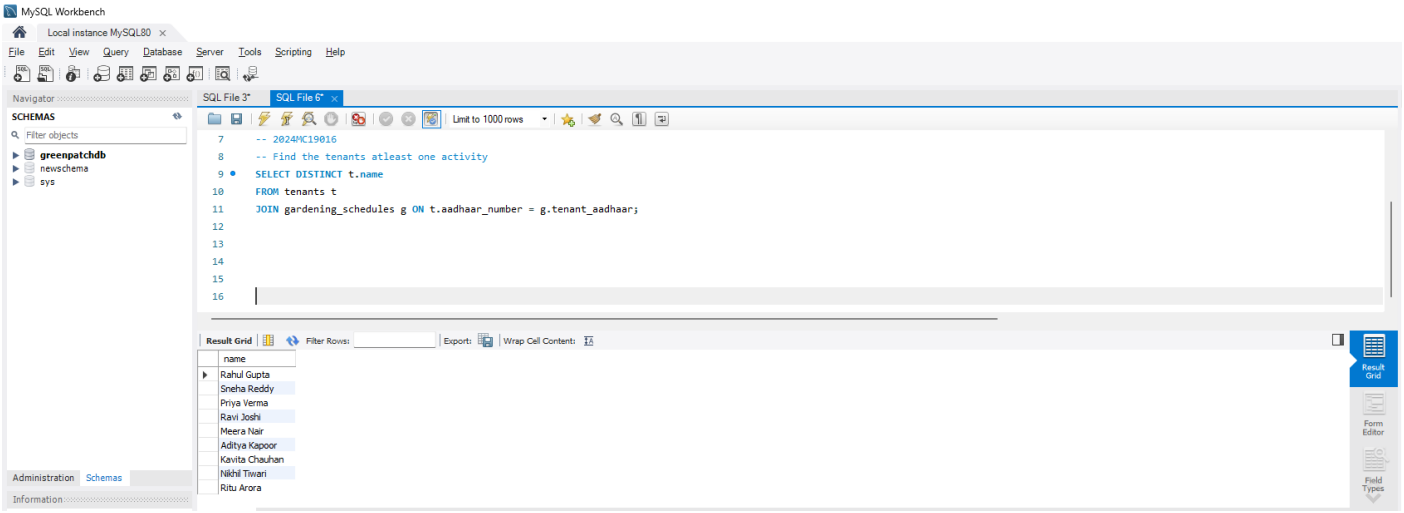


2024MC19016 Sankarraj V (Tenant Manager)

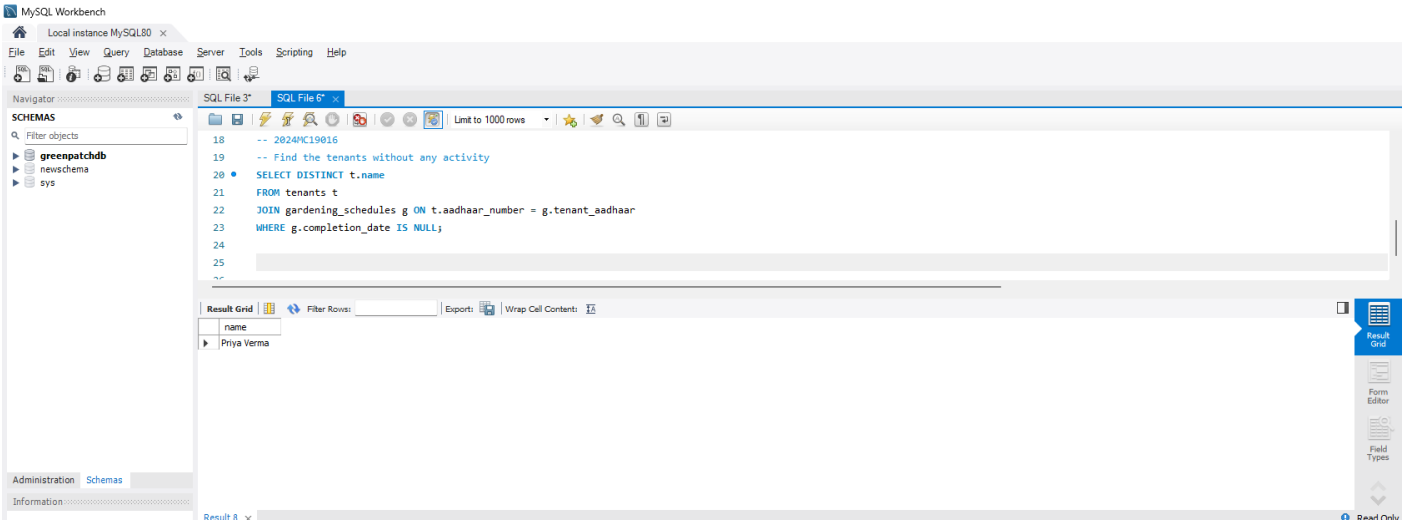
1.) List all the tenants and their count



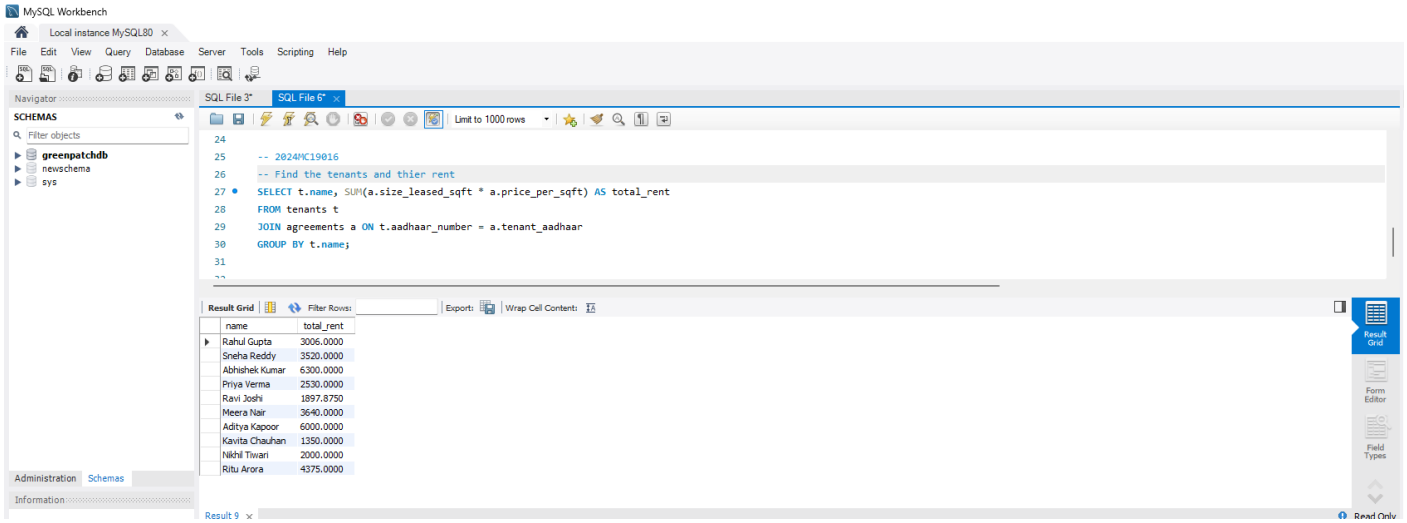
2.) Find the tenants with atleast one activity



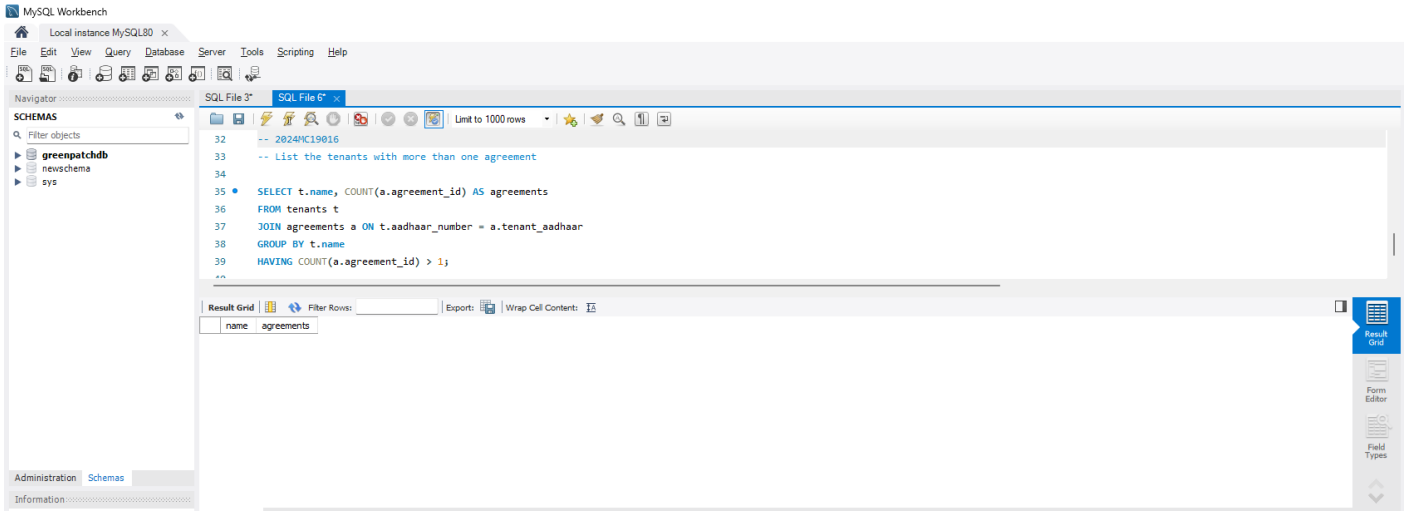
3.) Find the tenants without any activity



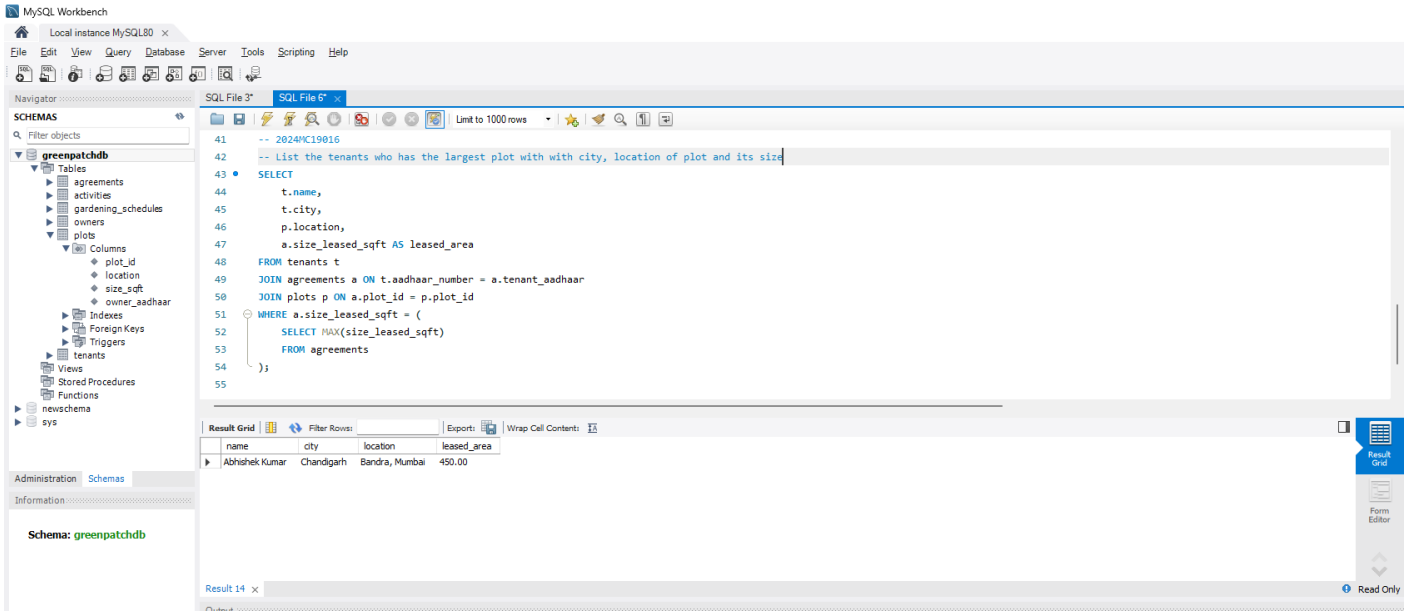
4.) Find the tenants and their rent



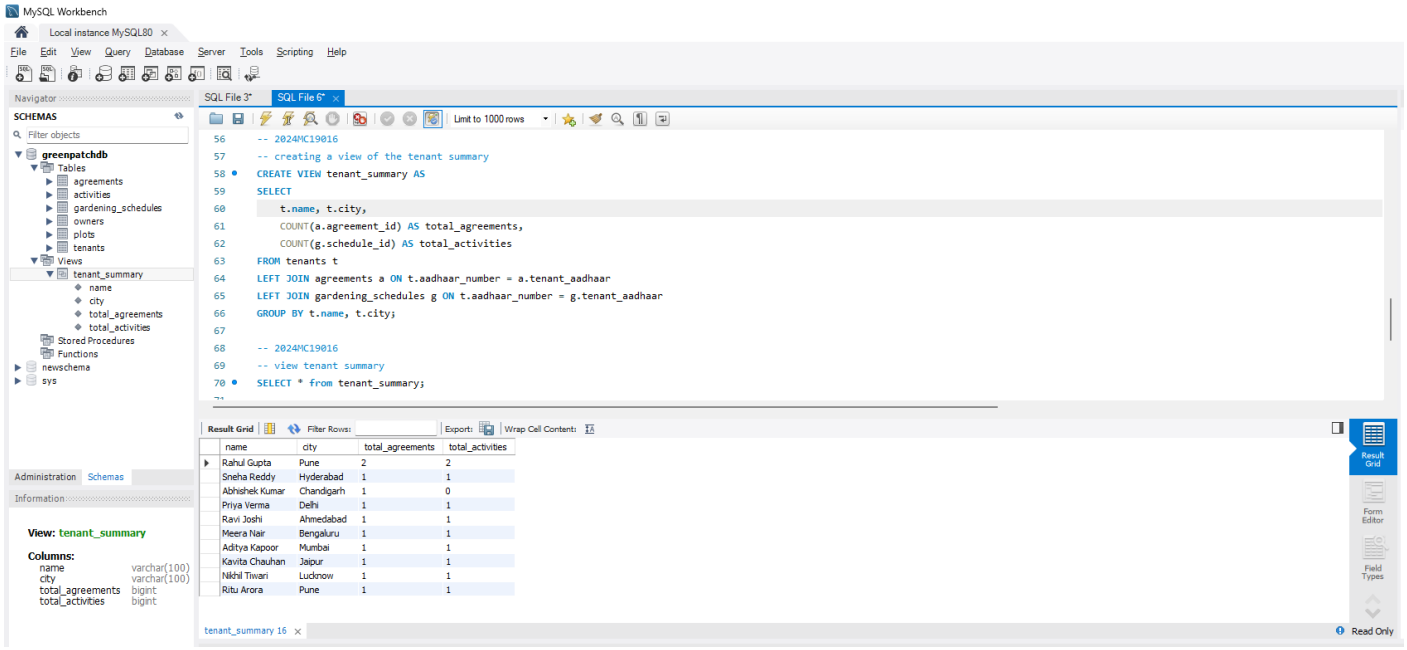
5.) List the tenants with more than one agreement



6.) List the tenants who has the largest plot with city, location and its size

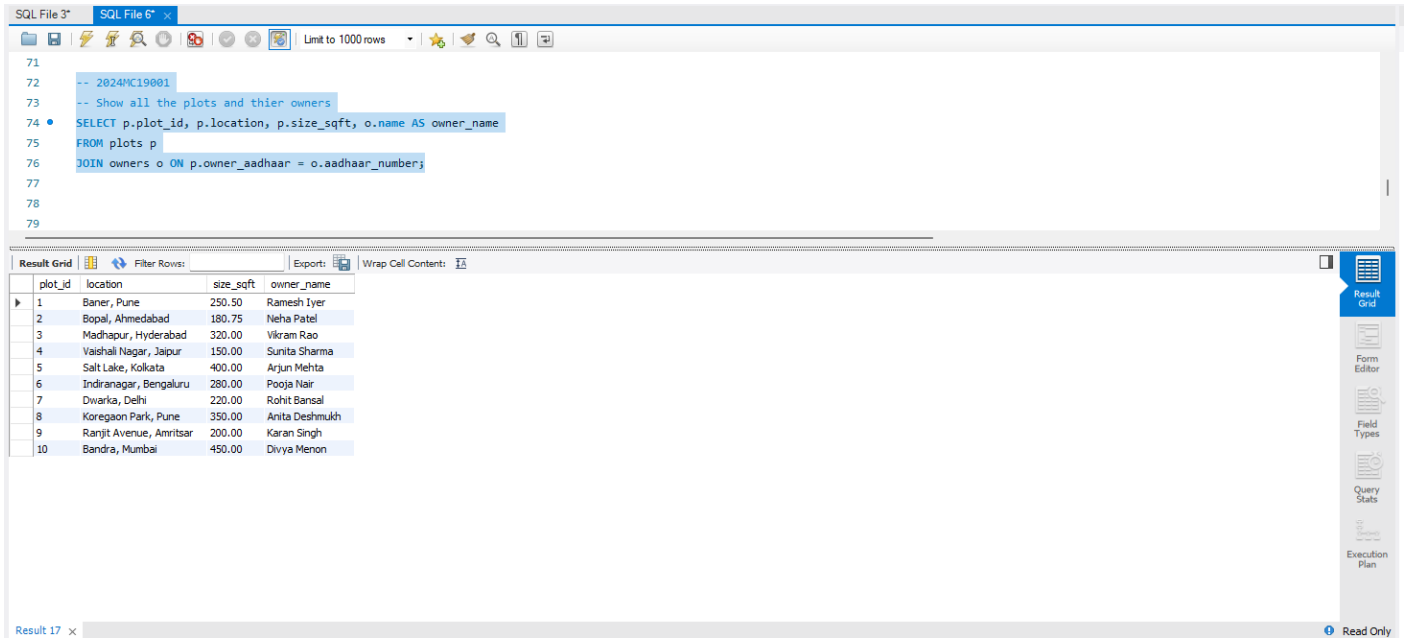


7.) Creating a view of the tenant summary and viewing the summary



2024MC19006 Ajith B (Owner Manager)

8.) Show all the plots and their owners



9.) Show owner and total plot they own

```
71
72      -- 2024MC19001
73      -- Show all the plots and thier owners
74      SELECT p.plot_id, p.location, p.size_sqft, o.name AS owner_name
75      FROM plots p
76      JOIN owners o ON p.owner_aadhaar = o.aadhaar_number;
77
78      -- 2024MC19001
79      -- show owner and total plots they own
```

name	total_plots
Ramesh Iyer	1
Neha Patel	1
Vikram Rao	1
Sunita Sharma	1
Arjun Mehta	1
Pooja Nair	1
Rohit Bansal	1
Anita Deshmukh	1
Karan Singh	1
Divya Menon	1

10.)Average plot size per owner

```
84
85      -- 2024MC19001
86      -- Average plot size per owner
87      SELECT o.name, AVG(p.size_sqft) AS avg_plot_size
88      FROM owners o
89      JOIN plots p ON o.aadhaar_number = p.owner_aadhaar
90      GROUP BY o.name;
91
92
```

name	avg_plot_size
Ramesh Iyer	250.500000
Neha Patel	180.750000
Vikram Rao	320.000000
Sunita Sharma	150.000000
Arjun Mehta	400.000000
Pooja Nair	280.000000
Rohit Bansal	220.000000
Anita Deshmukh	350.000000
Karan Singh	200.000000
Divya Menon	450.000000

11.) Find owner with no active agreement

```
91
92      -- 2024MC19001
93      -- Find owner with no active agreement
94      SELECT o.name
95      FROM owners o
96      LEFT JOIN agreements a ON o.aadhaar_number = a.owner_aadhaar
97      WHERE a.agreement_id IS NULL;
98
99
```

name

12.) List owners whose plot have completed gardening activities

```sql
-- 2024MC19001
-- List owners whose plot have completed gardening activities
SELECT DISTINCT o.name
FROM owners o
WHERE o.aadhaar_number IN (
    SELECT p.owner_aadhaar
    FROM plots p
    JOIN gardening_schedules g ON p.plot_id = g.plot_id
    WHERE g.completion_date IS NOT NULL
);
```

name
Ramesh Iyer
Neha Patel
Vikram Rao
Sunita Sharma
Arjun Mehta
Pooja Nair
Anita Deshmukh
Karan Singh

2024MC19015 Santhosh (Gardening Supervisor)

13.) List all scheduled gardening activities

```sql
-- 2024MC19015
-- List all scheduledgardening activities
SELECT g.schedule_id, a.activity_name, g.scheduled_date, g.completion_date
FROM gardening_schedules g
JOIN activities a ON g.activity_id = a.activity_id
ORDER BY g.scheduled_date;
```

schedule_id	activity_name	scheduled_date	completion_date
1	Soil Preparation	2025-01-05	2025-01-06
2	Planting	2025-01-08	2025-01-09
3	Watering	2025-02-10	2025-02-10
4	Weeding	2025-03-20	2025-03-21
5	Fertilizing	2025-04-15	2025-04-16
6	Pest Control	2025-05-20	2025-05-22
7	Harvesting	2025-06-25	2025-06-27
8	Mulching	2025-07-10	NULL
9	Composting	2025-08-18	2025-08-19
10	Pruning	2025-09-22	2025-09-23

14.)Show activities pending completion

SQL File 3* SQL File 6*

Limit to 1000 rows

```
-- 2024MC19015
-- Show activities pending completion
SELECT g.schedule_id, a.activity_name
FROM gardening_schedules g
JOIN activities a ON g.activity_id = a.activity_id
WHERE g.completion_date IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

schedule_id	activity_name
8	Mulching

15.) Count total activities per tenant

SQL File 3* SQL File 6*

Limit to 1000 rows

```
-- 2024MC19015
-- Count total activities per tenant
SELECT t.name, COUNT(g.schedule_id) AS total_activities
FROM tenants t
JOIN gardening_schedules g ON t.aadhaar_number = g.tenant_aadhaar
GROUP BY t.name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	total_activities
Rahul Gupta	2
Sneha Reddy	1
Priya Verma	1
Ravi Joshi	1
Meera Nair	1
Aditya Kapoor	1
Kavita Chauhan	1
Nikhil Tiwari	1
Ritu Arora	1

16.) Get activity count by type

```sql
-- 2024MC19015
-- Get activity count by type
SELECT a.activity_name, COUNT(g.schedule_id) AS count
FROM activities a
JOIN gardening_schedules g ON a.activity_id = g.activity_id
GROUP BY a.activity_name;
```

activity_name	count
Soil Preparation	1
Planting	1
Watering	1
Weeding	1
Fertilizing	1
Pest Control	1
Harvesting	1
Mulching	1
Composting	1
Pruning	1

17.) Display tenants and their completed activities

```sql
-- 2024MC19015
-- Display tenants and thier total completed activities
SELECT t.name, COUNT(g.schedule_id) AS completed_activities
FROM tenants t
JOIN gardening_schedules g ON t.aadhaar_number = g.tenant_aadhaar
WHERE g.completion_date IS NOT NULL
GROUP BY t.name;
```

name	completed_activities
Rahul Gupta	2
Ravi Joshi	1
Sneha Reddy	1
Kavita Chauhan	1
Aditya Kapoor	1
Meera Nair	1
Ritu Arora	1
Nikhil Tiwari	1

2024MC19001 Deeksha B (Agreement Manager)

18.) Display all agreements with tenant and owner details

```
49    #2024MC19001
50    #1. Display all agreements with tenant and owner details.
51 •  SELECT a.agreement_id, t.name AS tenant_name, o.name AS owner_name,
52           a.start_date, a.end_date
53    FROM agreements a
54    JOIN tenants t ON a.tenant_aadhaar = t.aadhaar_number
55    JOIN owners o ON a.owner_aadhaar = o.aadhaar_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

agreement_id	tenant_name	owner_name	start_date	end_date
1	Rahul Gupta	Ramesh Iyer	2025-01-01	2025-12-31
2	Ravi Joshi	Neha Patel	2025-02-01	2026-01-31
3	Sneha Reddy	Vikram Rao	2025-03-10	2026-03-09
4	Kavita Chauhan	Sunita Sharma	2025-04-01	2026-03-31
5	Aditya Kapoor	Arjun Mehta	2025-05-01	2026-04-30
6	Meera Nair	Pooja Nair	2025-06-15	2026-06-14
7	Priya Verma	Rohit Bansal	2025-07-01	2026-06-30
8	Ritu Arora	Anita Deshmukh	2025-08-01	2026-07-31
9	Nikhil Tiwari	Karan Singh	2025-09-10	2026-09-09
10	Abhishek Kumar	Divya Menon	2025-10-01	2026-09-30

19.) Find plots rented by each tenant

```
57    #2024MC19001
58    #2. Find plots rented by each tenant.
59 •  SELECT t.name, p.location
60    FROM tenants t
61    JOIN agreements a ON t.aadhaar_number = a.tenant_aadhaar
62    JOIN plots p ON a.plot_id = p.plot_id
63    WHERE p.size_sqft > 200;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	location
Rahul Gupta	Baner, Pune
Sneha Reddy	Madhapur, Hyderabad
Aditya Kapoor	Salt Lake, Kolkata
Meera Nair	Indiranagar, Bengaluru
Priya Verma	Dwarka, Delhi
Ritu Arora	Koregaon Park, Pune
Abhishek Kumar	Bandra, Mumbai

20.) Get agreement details expiring this year

```
78    #2024MC19001
79    #3. Get agreement details expiring this year.
80 •  SELECT *
81    FROM agreements
82    WHERE end_date BETWEEN '2025-01-01' AND '2025-12-31';
83
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

agreement_id	owner_aadhaar	tenant_aadhaar	plot_id	start_date	end_date	size_leased_sqft	price_per_sqft
1	510022334455	620033445566	1	2025-01-01	2025-12-31	250.50	12.00

21.) Calculate total rent (area x price)

```
84      #2024MC19001
85      #4. Calculate total rent (area x price).
86 •    SELECT a.agreement_id, (a.size_leased_sqft * a.price_per_sqft) AS total_rent
87      FROM agreements a;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

agreement_id	total_rent
1	3006.0000
2	1897.8750
3	3520.0000
4	1350.0000
5	6000.0000
6	3640.0000
7	2530.0000
8	4375.0000
9	2000.0000
10	6300.0000

22.) Show plots leased for more than 6 months

```
89      #2024MC19001
90      #5. Show plots leased for more than 6 months.
91 •    SELECT a.agreement_id, t.name, DATEDIFF(a.end_date, a.start_date) AS duration
92      FROM agreements a
93      JOIN tenants t ON a.tenant_aadhaar = t.aadhaar_number
94      WHERE DATEDIFF(a.end_date, a.start_date) > 180;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

agreement_id	name	duration
1	Rahul Gupta	364
2	Ravi Joshi	364
3	Sneha Reddy	364
4	Kavita Chauhan	364
5	Aditya Kapoor	364
6	Meera Nair	364
7	Priya Verma	364
8	Ritu Arora	364
9	Nikhil Tiwari	364
10	Abhishek Kumar	364

23.) Show agreement which has the highest total value

```
96    #2024MC19001
97    #6. Show Agreement which has the highest total value
98    SELECT
99        agreement_id,
100       owner_aadhaar,
101       tenant_aadhaar,
102       (size_leased_sqft * price_per_sqft) AS total_value
103   FROM Agreements
104   ORDER BY total_value DESC
105   LIMIT 1;
106
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

agreement_id	owner_aadhaar	tenant_aadhaar	total_value
10	510022334464	620033445568	6300.0000

24.) The monthly rent per agreement

```
107   #2024MC19001
108   #7. The monthly rent per agreement
109   SELECT
110       agreement_id,
111       ROUND((size_leased_sqft * price_per_sqft) / TIMESTAMPDIFF(MONTH, start_date, end_date), 2)
112       AS monthly_rent
113   FROM Agreements;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

agreement_id	monthly_rent
1	273.27
2	172.53
3	320.00
4	122.73
5	545.45
6	330.91
7	230.00
8	397.73
9	181.82
10	572.73

2024MC19005 Sheetal (Admin)

25.) Find the plots with no gardening activities.

```
#2024MC19005
#1. Find plots with no gardening activities.
SELECT p.plot_id, p.location
FROM plots p
LEFT JOIN gardening_schedules g ON p.plot_id = g.plot_id
WHERE g.plot_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plot_id	location
10	Bandra, Mumbai

26.) Find the most expensive agreement per square foot.

```
#2024MC19005
#2. Find the most expensive agreement per square foot.
SELECT *
FROM agreements
ORDER BY price_per_sqft DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

agreement_id	owner_aadhaar	tenant_aadhaar	plot_id	start_date	end_date	size_leased_sqft	price_per_sqft
5	510022334459	620033445572	5	2025-05-01	2026-04-30	400.00	15.00

27.) Find the owner with the largest total area owned.

```
#2024MC19005
#3. Find the owner with the largest total area owned.
SELECT o.name, SUM(p.size_sqft) AS total_area
FROM owners o
JOIN plots p ON o.aadhaar_number = p.owner_aadhaar
GROUP BY o.name
ORDER BY total_area DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	total_area
Divya Menon	450.00

28.) Combine all tenant and owner names.

```
28      #2024MC19005
29      # 4. Combine all tenant and owner names.
30 •    SELECT o.Name AS Person_Name,'Owner' AS Role FROM OWNERS o
31      UNION
32      SELECT t.Name AS Person_Name,'Tenant' AS Role FROM TENANTS t
33      ORDER BY Role, Person_Name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Person_Name	Role
Anita Deshmukh	Owner
Arjun Mehta	Owner
Divya Menon	Owner
Karan Singh	Owner
Neha Patel	Owner
Pooja Nair	Owner
Ramesh Iyer	Owner
Rohit Bansal	Owner
Sunita Sharma	Owner
Vikram Rao	Owner
Abhishek Kumar	Tenant
Aditya Kapoor	Tenant
Kavita Chauhan	Tenant
Meera Nair	Tenant
Nikhil Tiwari	Tenant
Priya Verma	Tenant

29.) Create a view for most active tenants.

```
36      #5. Create a view for most active tenants.
37 •    CREATE VIEW tenant_activity AS
38      SELECT t.name, COUNT(g.schedule_id) AS task_count
39      FROM tenants t
40      JOIN gardening_schedules g ON t.aadhaar_number = g.tenant_aadhaar
41      GROUP BY t.name;
42 •    SELECT *
43      FROM tenant_activity_summary;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	task_count
Rahul Gupta	2
Sneha Reddy	1
Priya Verma	1
Ravi Joshi	1
Meera Nair	1
Aditya Kapoor	1
Kavita Chauhan	1
Nikhil Tiwari	1
Ritu Arora	1

```
35      #2024MC19005
36      #5. Create a view for most active tenants.
37 •    CREATE VIEW tenant_activity AS
38      SELECT t.name, COUNT(g.schedule_id) AS task_count
39      FROM tenants t
40      JOIN gardening_schedules g ON t.aadhaar_number = g.tenant_aadhaar
41      GROUP BY t.name;
42 •    SELECT *
43      FROM tenant_activity_summary;
44 •    SELECT *
45      FROM tenant_activity_summary
46      ORDER BY task_count DESC
47      LIMIT 1;
48
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	task_count
Rahul Gupta	2

```
49    #2024MC19005
50    #6. Convert location as atomic
51 •  ALTER TABLE Plots
52    ADD COLUMN Area VARCHAR(50),
53    ADD COLUMN City VARCHAR(50);
54 •  SET SQL_SAFE_UPDATES = 0;
55 •  UPDATE Plots
56    SET
57        Area = TRIM(SUBSTRING_INDEX(Location, ',', 1)),
58        City = TRIM(SUBSTRING_INDEX(Location, ',', -1));
59 •  SET SQL_SAFE_UPDATES = 1;
60 •  SELECT PlotID, location,Area, City FROM Plots;
61
```

Output

Action Output

#	Time	Action	Message
50	10:26:58	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
51	10:27:03	UPDATE Plots SET Area = TRIM(SUBSTRING_INDEX(Location, ',', 1)), City ...	10 row(s) affected Rows matched: 10 Changed: 10 Warnings: 0
52	10:27:05	SET SQL_SAFE_UPDATES = 1	0 row(s) affected

30.) Convert location as atomic.

```
49    #2024MC19005
50    #6. Convert location as atomic
51 •  ALTER TABLE Plots
52    ADD COLUMN Area VARCHAR(50),
53    ADD COLUMN City VARCHAR(50);
54 •  SET SQL_SAFE_UPDATES = 0;
55 •  UPDATE Plots
56    SET
57        Area = TRIM(SUBSTRING_INDEX(Location, ',', 1)),
58        City = TRIM(SUBSTRING_INDEX(Location, ',', -1));
59 •  SET SQL_SAFE_UPDATES = 1;
60 •  SELECT * FROM Plots;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

plot_id	location	size_sqft	owner_aadhaar	Area	City
1	Baner, Pune	250.50	510022334455	Baner	Pune
2	Bopal, Ahmedabad	180.75	510022334456	Bopal	Ahmedabad
3	Madhapur, Hyderabad	320.00	510022334457	Madhapur	Hyderabad
4	Vaishali Nagar, Jaipur	150.00	510022334458	Vaishali Nagar	Jaipur
5	Salt Lake, Kolkata	400.00	510022334459	Salt Lake	Kolkata
6	Indiranagar, Bengaluru	280.00	510022334460	Indiranagar	Bengaluru
7	Dwarka, Delhi	220.00	510022334461	Dwarka	Delhi
8	Koregaon Park, Pune	350.00	510022334462	Koregaon Park	Pune
9	Ranjit Avenue, Amritsar	200.00	510022334463	Ranjit Avenue	Amritsar
10	Bandra, Mumbai	450.00	510022334464	Bandra	Mumbai